

MESOSCALE ANALYSIS PARAMETERS

(Material Excerpted From SPC Mesoanalysis Page)

Level of Free Convection (LFC)

The Level of Free Convection (LFC) is the level at which a lifted parcel begins a free acceleration upward to the equilibrium level. Recent preliminary research suggests that tornadoes become more likely in supercells when LFC heights are less than 2000 m (6500 feet) above ground level. The EL (equilibrium level) is the level at which a lifted parcel becomes cooler than the environmental temperature and is no longer buoyant (i.e., "unstable"). The EL is used primarily to estimate the height of a thunderstorm anvil. The height difference between this parameter and the LCL is important when determining convection initiation. The smaller the difference between the LFC and the LCL, the more likely deep convection becomes. The LFC-LCL difference is similar to CIN (convective inhibition).

Lifting Condensation Level (LCL)

The Lifting Condensation Level (LCL) is the level at which a parcel becomes saturated. It is a reasonable estimate of cloud base height when parcels experience forced ascent. The height difference between this parameter and the LFC is important when determining convection initiation. The smaller the difference between the LCL and the LFC, the more likely deep convection becomes. The LFC-LCL difference is similar to CIN (convective inhibition). LCL heights from approximately 500 m (1600 ft) to 800 m (2600 ft) above ground level are associated with F2 to F5 tornadoes. Low LCL heights and low surface dewpoint depressions (high low level RH) suggest a warm RFD which may play a role in tornado development.

Convective Condensation Level (CCL)

The Convective Condensation Level (CCL) is the level at which condensation will occur if sufficient afternoon heating causes rising parcels of air to reach saturation. The CCL is greater than or equal in height (lower or equal pressure level) than the LCL. The CCL and the LCL are equal when the atmosphere is saturated. The CCL is found at the intersection of the saturation mixing ratio line (through the surface dewpoint) and the environmental temperature.

Equilibrium Level (EL)

The Equilibrium Level (EL) is the level at which a lifted parcel becomes cooler than the environmental temperature and is no longer buoyant (i.e. unstable). The EL is used primarily to estimate the height of a thunderstorm anvil. You may notice that the "virtual" and "non-virtual" lifted parcels both end up with the same EL. This happens because the virtual temperature converges to the actual temperature when temperatures are very cold (less than -20C) and moisture effects become negligible.

Lapse Rates (C/km)

A lapse rate is the rate of temperature change with height. The faster the temperature decreases with height, the "steeper" the lapse rate and the more "unstable" the atmosphere becomes. Lapse rates are typically displayed in ranges from 850-500-mb (4,500-18,000-ft above sea level) and 700-500-mb (10,000-18,000-ft above sea level).

Lapse rates are shown in terms of degrees Celsius change per kilometer in height. Values less than 5.5-6.0 degrees C/km ("moist" adiabatic) represent "stable" conditions, while values near 9.5 degrees C/km ("dry" adiabatic) are considered "absolutely unstable." In between these two values, lapse rates are considered "conditionally unstable." Conditional instability means that if enough moisture is present, lifted air parcels could have a negative LI (lifted index) or positive CAPE.

Surface-Based CAPE (SBCAPE - J/kg)

SBCAPE (Surface-Based Convective Available Potential Energy) is a measure of instability in the troposphere. This value represents the total amount of potential energy available to a parcel of air originating at the surface and being lifted to its level of free convection (LFC). No parcel entrainment is considered. The CAPE calculation uses virtual temperatures, but the CIN value does not. SPC forecasters have noted that non-virtual CIN calculations tend to define areas of weak cap more accurately than if the virtual temperature was considered.

Convective Inhibition (CIN - J/kg)

CIN (Convective Innhibition) Represents the "negative" area on a sounding that must be overcome before storm initiation can occur.

Surface-Based Lifted Index (SBLI - C) & Convective Inhibition (CIN - J/kg)

SBLI (**S**urface **B**ased **L**ifted **I**ndex & **C**onvective **I**nhibition) is the Lifted Index at 500-mb, based on the most unstable parcel, and the convective inhibition for the same parcel. These fields are meant to identify areas of surface-based CAPE and minimal convective inhibition, which suggests some threat for surface-based thunderstorms.

Most Unstable CAPE (MUCAPE - J/kg)

MUCAPE (**M**ost **U**nstable **C**onvective **A**vailable **P**otential **E**nergy) is a measure of instability in the troposphere. This value represents the total amount of potential energy available to the most unstable parcel of air found within the lowest 300-mb of the atmosphere while being lifted to its level of free convection (LFC). No parcel entrainment is considered. The CAPE calculation uses virtual temperatures, but the CIN value does not. SPC forecasters have noted that non-virtual CIN calculations tend to define areas of weak cap more accurately than if the virtual temperature was considered.

LPL Height (m AGL)

The LPL (**L**ifted **P**arcel **L**evel) allows for the determination of the height of the most unstable parcel. This makes it easy to identify areas where the largest CAPE is "elevated."

100-mb Mixed Layer CAPE/CIN (J/kg)

MLCAPE (**M**ixed **L**ayer **C**onvective **A**vailable **P**otential **E**nergy) is a measure of instability in the troposphere. This value represents the mean potential energy conditions available to parcels of air located in the lowest 100-mb when lifted to the level of free convection (LFC). No parcel entrainment is considered. The CAPE calculation uses virtual temperatures, but the CIN value does not. SPC forecasters have noted that non-virtual CIN calculations tend to define areas of weak cap more accurately than if the virtual temperature was considered.

3-km CAPE (J/kg) & Surface Vorticity

CAPE in the lowest 3-km above ground level, and surface relative vorticity. Areas of large 0-3-km CAPE tend to favor strong low-level stretching, and can support tornado formation when co-located with significant vertical vorticity near the ground.

Normalized CAPE (J/kg)

The NCAPE (**N**ormalized **C**APE) is CAPE that is divided by the depth of the buoyancy layer (units of m s^{-2}). Values near or less than .1 suggest a "tall, skinny" CAPE profile with relatively weak parcel accelerations, while values closer to .3 to .4 suggest a "fat" CAPE profile with large parcel accelerations possible. Normalized CAPE and lifted indices are similar measures of instability.

Downdraft CAPE (J/kg)

The DCAPE (**D**owndraft **C**APE) can be used to estimate the potential strength of rain-cooled downdrafts within thunderstorm convection, and is similar to CAPE. Larger DCAPE values are associated with stronger downdrafts.

Surface to 6-km Vertical Shear Vector (kts)

The Boundary Layer through 6-km above ground level shear vector denotes the change in wind throughout this height. Thunderstorms tend to become more organized and persistent as vertical shear increases. Supercells are commonly associated with vertical shear values of 35-40 knots and greater through this depth.

Effective Bulk Vertical Shear Vector (kts)

The maximum bulk shear from the most unstable parcel level upward to 40-60% of the equilibrium level height. This parameter is similar to the 0-6 km bulk shear, though it accounts for storm depth (LPL to EL) and is designed to identify both surface-based and "elevated" supercell environments. Supercells become more probable as the effective bulk shear increases through the range of 25-40 kt and greater.

Bulk Richardson Number Shear (m^2/s^2)

The BRN (**B**ulk **R**ichardson **N**umber) shear is similar to the BL-6-km shear, except that the BRN Shear uses a difference between the low-level wind and a density-weighted mean wind through the mid-levels. Values of 35-40 m^2/s^2 or greater have been associated with supercells.

Storm Relative Helicity (m^2/s^2)

SRH (**S**torm **R**elative **H**elicity) is a measure of the potential for cyclonic updraft rotation in right-moving supercells, and is calculated for the lowest 1-km and 3-km layers above ground level. There is no clear threshold value for SRH when forecasting supercells, since the formation of supercells appears to be related more strongly to the deeper layer vertical shear. Larger values of 0-3-km SRH (greater than 250 m^2/s^2) and 0-1-km SRH (greater than 100 m^2/s^2), however, do suggest an increased threat of tornadoes with supercells. For SRH, larger values are generally better, but there are no clear "boundaries" between non-tornadic and significant tornadic supercells.

Effective Storm Relative Helicity (m^2/s^2)

Effective SRH (**S**torm **R**elative **H**elicity) is based on threshold values of lifted parcel CAPE (100 J kg^{-1}) and CIN (-250 J kg^{-1}). These parcel constraints are meant to confine the SRH layer calculation to the part of a sounding where lifted parcels are

buoyant, but not strongly capped. For example, a supercell forms or moves over an area where the most unstable parcels are located a couple of thousand feet above the ground, and stable air is located at ground level. The question then becomes "how much of the cool air can the supercell ingest and still survive?" Our estimate is to start with the surface parcel level, and work upward until a lifted parcel's CAPE value increases to 100 J kg^{-1} or more, with an associated CIN greater than -250 J kg^{-1} . From the level meeting the constraints (the "effective surface"), we continue to look upward in the sounding until a lifted parcel has a CAPE less than 100 J kg^{-1} OR a CIN less than -250 J kg^{-1} . Of the three SRH calculations displayed on the SPC mesoanalysis page, effective SRH discriminates the best between significant tornadic and nontornadic supercells.

Surface-1-km Vertical Shear Vector (kts)

Surface-1-km Vertical Shear is the difference between the surface wind and the wind at 1-km above ground level. These data are plotted as vectors with shear magnitudes contoured. 0-1-km shear magnitudes greater than 15-20 knots tend to favor supercell tornadoes.

Surface-2-km Storm Relative Winds (kts)

Low-Level SR (**S**torm **R**elative) winds (0-2-km) are meant to represent low-level storm inflow. The majority of sustained supercells have 0-2-km storm inflow values of 15-20 knots or greater.

4-6-km Storm Relative Winds (kts)

Mid-Level SR (**S**torm **R**elative) winds (4-6-km) are of some use in discriminating between tornadic and non-tornadic supercells. Tornadic supercells tend to have 4-6-km SR wind speeds in excess of 15 knots, while non-tornadic supercells tend to have weaker mid-level storm-relative winds.

Anvil Level/9-11-km SR Winds (kts)

The Anvil Level SR (**S**torm **R**elative) winds and SR winds from 9-11-km are meant to discriminate supercell type. In general, upper-level SR winds less than 40 knots correspond to "high precipitation" supercells, 40-60 knots SR winds denote "classic" supercells, while SR winds greater than 60 knots correspond to "low precipitation" supercells.

Supercell Composite Parameter

A multi-parameter index that includes effective SRH, muCAPE, and effective bulk shear. Each parameter is normalized to supercell "threshold" values. Effective SRH is divided by $50 \text{ m}^2/\text{s}^2$, muCAPE is divided by 1000 J/kg , and effective bulk shear is divided by 20 m/s in the shear range of 10-20 m/s. Effective bulk shear less than 10 m/s is set to zero, and effective bulk shear greater than 20 m/s is set to one.

Significant Tornado Parameter

A multi-parameter index that includes effective bulk shear, effective SRH, 100-mb mean parcel CAPE, 100-mb mean parcel CIN, and 100-mb mean parcel LCL height. When the mLCL is less than 1000 m AGL, the mLCL term is set to one, and when the mL CIN is greater than -50 J kg^{-1} , the mL CIN term is set to one. Lastly, the ESHEAR term is capped at a value of 1.5, and set to zero when ESHEAR is less than 12.5 m s^{-1} . A majority of significant tornadoes (F2 or greater damage) have been associated with STP values greater than 1, while most non-tornadic supercells have been associated with values less than 1 in a large sample of RUC analysis proximity soundings.

Significant Hail Parameter

The Sig. Hail Parameter (SHIP) was developed using a large database of surface-modified, observed severe hail proximity soundings. It is based on 5 parameters, and is meant to delineate between SIG ($\geq 2"$ diameter) and NON-SIG ($< 2"$ diameter) hail environments. It is important to note that **SHIP is NOT a forecast hail size**. Since SHIP is based on the RUC depiction of MUCAPE, unrepresentative MUCAPE "bullseyes" may cause a similar increase in SHIP values. This typically occurs when bad surface observations get into the RUC model. Developed in the same vein as the STP and SCP parameters, values of SHIP greater than 1.00 indicate a favorable environment for SIG hail. Values greater than 4 are considered very high. In practice, maximum contour values of 1.5-2.0 or higher will typically be present when SIG hail is going to be reported.

Craven SigSvr Parameter

The simple product of 100mb MLCAPE and 0-6km magnitude of the vector difference (m/s; often referred to as "deep layer shear") accounts for the compensation between instability and shear magnitude. Using a database of about 60,000 soundings, the majority of significant severe events (2+ inch hail, 65+ knot winds, F2+ tornadoes) occur when the product exceeds 20,000 m^3/s^3 . For example, a 0-6-km shear of 40 knots and CAPE of 3000 J/kg results in a Craven SigSvr index of 60,000. Units are scaled to the nearest 1000 on the web plot.

Energy-Helicity Index

The basic premise behind the EHI (**E**nergy-**H**elicity **I**ndex) is that storm rotation should be maximized when CAPE is large and SRH is large. 0-1-km EHI values greater than 1-2 have been associated with significant tornadoes in supercells.

Vorticity Generation Parameter (m/s^2)

The VGP **V**orticity **G**eneration **P**arameter) is meant to estimate the rate of tilting and stretching of horizontal vorticity by a thunderstorm updraft. Values greater than 0.2 m/s^2 suggest an increasing possibility of tornadic storms.